

Pennada Venkata Chaitanya

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Professional Summary

AI/ML Software Engineer with **2.5+ years of experience** building production GenAI systems, AI agents, distributed backend systems, and scalable data pipelines. Delivered up to **12× faster processing** across high-throughput production workflows using Python and AI-driven automation.

Work Experience

Associate Software Engineer | Innocito

April 2025 - Present

- Built **distributed data pipelines** across parallel workers using Python, Prefect, FastAPI, and MongoDB, with **lease-based coordination and asynchronous processing**, cutting processing time from **14 days to 1 day**.
- Processed and mapped **5 million+ product records** through high-throughput AI pipelines for large-scale data preprocessing, classification, and catalog enrichment.
- Built a real-time **AI Voice Agent** integrating LLMs, speech-to-text, and text-to-speech, reducing end-to-end response latency by **3 seconds per request** for more natural real-time conversations.

AI Software Developer | Freelance

March 2024 - February 2025

- Developed AI-driven text-to-video workflows, achieving **80% visual consistency** across predefined test cases.
- Built an automated AI GIF Studio, achieving **95% test-case accuracy, 5× faster runs, and 10,000+ interactions**.
- Developed AI agents for social media automation, reducing manual task time by **30%**.

AI/ML SDE Intern | Spring Works

July 2023 - September 2023

- Developed LLM-based POCs and automated unit testing with Pytest, increasing test coverage from **30% to 75%**.

Projects

Synapse

[Website](#) | [GitHub](#)

- Architected an **offline-first AI knowledge system** combining block-based knowledge management, information retrieval, task orchestration, and agent-driven actions.
- Designed **distributed synchronization** between SQLite and PostgreSQL with block-level incremental updates and HLC-based conflict resolution.
- Built **hybrid local-cloud RAG pipelines** with incremental vector indexing, retrieval, and MCP-based tool execution.
Tech: Python, FastAPI, PostgreSQL, LangGraph, Ollama, MCP.

Ed AI

[GitHub](#)

- Built an AI learning assistant featuring course recommendations, Socratic tutoring, and AI-powered mock interviews.
Tech: Python, FastAPI, LangChain, RAG, Streamlit.

Skills

Programming: Python, SQL, C++

Generative AI & Agents: GenAI, AI Agents, RAG, Multimodal LLMs, LangGraph, LangChain, LlamaIndex, MCP, Vector Search, Ollama

ML & AI: PyTorch, NLP, Computer Vision, Model Evaluation, Data Preprocessing, Feature Engineering

Backend & Data: FastAPI, PostgreSQL, MongoDB (NoSQL), SQLite, Prefect, REST APIs

Engineering Tools: Pytest, Playwright, Git, GitHub, Jira, Agile/Scrum, Automation, Streamlit, Docker

Achievements

- Won the institutional round of the **Solving for India Hackathon** hosted by Google, AMD, and GeeksforGeeks.
- Won the international **Fresh Hacks** hackathon hosted on [Devpost](#).
- Received the **Rising Star Award** at Innocito for strong performance, rapid growth, and technical contributions.
- Earned a **5-star Problem Solving** rating on [HackerRank](#). Solved **300+ DSA problems** across [LeetCode](#) & [GFG](#).

Education

B.Tech in Computer Science | Aditya College of Engineering and Technology

2020 - 2024

Focus: Artificial Intelligence & Machine Learning